

EXPLORATORY DATA ANALYSIS (EDA) PROJECT

Data Exploration, Statistical Analysis and Visualization

Dataset: Iris Dataset

Tools: Python, Pandas, Matplotlib, Seaborn, Scikit-learn

Project Report

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying advanced analytics or machine learning. It uses statistical summaries, data-quality checks, correlations, and visualizations to discover patterns, trends, relationships, and unusual observations. This project performs EDA on the Iris dataset using Python.

2. Objectives

The objectives are to understand dataset structure, calculate statistical summaries, check missing values, identify relationships between numerical variables, visualize distributions and relationships, and present meaningful observations in a structured report.

3. Tools and Technologies

The project was implemented in Google Colab using Python. Pandas was used for data handling, Scikit-learn for loading the Iris dataset, Matplotlib for plotting, and Seaborn for the correlation heatmap.

4. Dataset Description

The Iris dataset contains 150 observations and four numerical features: sepal length, sepal width, petal length, and petal width. It is a standard dataset used for learning data analysis and machine learning.

5. Methodology

The dataset was loaded using Scikit-learn and converted into a Pandas DataFrame. The first rows and dimensions were inspected. Descriptive statistics were calculated, missing values were checked, and a correlation matrix was created. Heatmap, scatter plot, histogram, and box plot visualizations were then generated.

6. Statistical Summary

The dataset contains 150 rows and 4 numerical columns. Descriptive statistics provide count, mean, standard deviation, minimum, quartiles, and maximum values for each feature.

7. Missing Value Analysis

The missing-value check was performed for every numerical feature. The Iris dataset contains no missing values in these analyzed features.

8. Correlation Analysis

Correlation measures the strength and direction of relationships between numerical variables. The correlation heatmap makes these relationships easy to identify. Petal measurements show particularly strong positive relationships.

9. Visualizations

The project produces a correlation heatmap, a scatter plot of sepal length versus petal length, a histogram of sepal length, and a box plot of the four numerical features.

10. Key Insights

The dataset contains 150 observations and 4 numerical features. There are no missing values. Petal length and petal width have a strong positive relationship. The scatter plot shows a positive relationship between sepal length and petal length. Histograms show feature distributions, while box plots show spread and possible unusual values.

11. Advantages of EDA

EDA helps analysts understand data before modeling. It can reveal missing values, unusual observations, distributions, relationships, and potentially important variables.

12. Applications

EDA is widely used in business analytics, healthcare, finance, marketing, customer analysis, education, scientific research, and machine learning.

13. Limitations

EDA describes patterns but does not prove causation. Correlation does not necessarily mean one variable causes another. Findings also depend on data quality and dataset size.

14. Future Scope

Future work can compare distributions across flower species, perform outlier analysis, apply PCA, and use the findings as preparation for classification models.

15. Conclusion

This project successfully demonstrates Exploratory Data Analysis using the Iris dataset. Statistical summaries, missing-value checks, correlation analysis, and visualizations were used to understand the data and identify important relationships.

16. Python Implementation

The following code was used to perform the EDA in Google Colab:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_iris

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

print("EXPLORATORY DATA ANALYSIS")
print(df.head())
print(df.shape)
print(df.describe())
print(df.isnull().sum())

correlation = df.corr()
print(correlation)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation, annot=True)
plt.title("Correlation Heatmap")
plt.show()

plt.figure(figsize=(8, 6))
plt.scatter(df["sepal length (cm)"], df["petal length (cm)"])
plt.xlabel("Sepal Length (cm)")
plt.ylabel("Petal Length (cm)")
plt.title("Sepal Length vs Petal Length")
plt.show()

plt.figure(figsize=(8, 6))
plt.hist(df["sepal length (cm)"], bins=10)
plt.title("Distribution of Sepal Length")
plt.show()

plt.figure(figsize=(8, 6))
plt.boxplot(df.values)
plt.title("Box Plot of Iris Features")
plt.show()
```

17. Project Summary

Parameter	Details
Project	Exploratory Data Analysis (EDA)
Dataset	Iris Dataset
Rows	150
Numerical Features	4
Missing Values	None
Analysis	Statistics, missing values, correlation
Visualizations	Heatmap, scatter plot, histogram, box plot
Platform	Google Colab

Submission note: Add your Colab output screenshots to the GitHub repository along with this PDF and the Python code.